

# Sequences and Series of Functions

Li Jiayi, Joanna

14 July, 2014

# Contents

|          |                                                           |          |
|----------|-----------------------------------------------------------|----------|
| <b>1</b> | <b>Review: Sequences and Series of Real Numbers</b>       | <b>4</b> |
| 1.1      | Sequences . . . . .                                       | 4        |
| 1.1.1    | Sequences . . . . .                                       | 4        |
| 1.1.2    | Some properties of a convergent sequence . . . . .        | 4        |
| 1.1.3    | Monotone Sequences . . . . .                              | 5        |
| 1.1.4    | Subsequences and the Bolzano-Weiertrass Theorem . . . . . | 6        |
| 1.1.5    | Cauchy's Convergence Criterion . . . . .                  | 7        |
| 1.2      | Series . . . . .                                          | 7        |
| 1.2.1    | Series . . . . .                                          | 7        |
| 1.2.2    | Some tests for convergence . . . . .                      | 8        |
| 1.2.3    | Absolute and Conditional Convergence . . . . .            | 8        |
| <b>2</b> | <b>Sequences and Series of Functions</b>                  | <b>9</b> |
| 2.1      | Pointwise convergence of sequences of functions . . . . . | 9        |
| 2.2      | Uniform convergence of sequences of functions . . . . .   | 11       |
| 2.2.1    | Uniform convergence and boundedness . . . . .             | 13       |
| 2.2.2    | Uniform convergence and continuity . . . . .              | 14       |
| 2.2.3    | Uniform convergence and Riemann integration . . . . .     | 16       |
| 2.2.4    | Uniform convergence and differentiation . . . . .         | 17       |

|       |                                                                     |    |
|-------|---------------------------------------------------------------------|----|
| 2.3   | Uniform convergence of series of functions . . . . .                | 20 |
| 2.3.1 | Some tests for uniform convergence of series of functions . . . . . | 20 |
| 2.3.2 | the Power Series . . . . .                                          | 20 |

# 1 Review: Sequences and Series of Real Numbers

## 1.1 Sequences

### 1.1.1 Sequences

**Definition 1.1.** A sequence of real numbers is a real-valued function defined on the set of natural numbers, i.e. a function  $f : \mathbb{N} \longrightarrow \mathbb{R}$ .

In other words, a sequence can be written as  $f(1), f(2), f(3), \dots$ . We shall denote by  $a_n$  such a sequence where  $a_n = f(n) \in \mathbb{R}$ . The number  $a_n$  is called the  $n$ -th term of the sequence.

**Definition 1.2.** A sequence  $\{a_n\}$  is said to converge to a real number  $L$  if  $\forall \epsilon > 0, \exists N \in \mathbb{N}$  such that if  $n > N$ , then  $|a_n - L| < \epsilon$ .

**Example 1.1.** Prove that the sequence  $\frac{n}{n+1}$  converges to 1, that is, prove that

$$\lim_{n \rightarrow \infty} \frac{n}{n+1} = 1$$

*Proof.* We are required to show that for any given  $\epsilon > 0$ , we can find a natural number  $N$  such that if  $n > N$ , then  $|\frac{n}{n+1} - 1| = |\frac{-1}{n+1}| < \epsilon$ .

We first note that  $\frac{1}{n+1} < \frac{1}{n}$  for any  $N \in \mathbb{N}$ . Now that for any given  $\epsilon$ , by the Archimedean Property, there exists  $N \in \mathbb{N}$  such that  $\frac{1}{N} < \epsilon$ . It follows that if  $n > N$ , then

$$\frac{1}{n+1} < \frac{1}{n} < \frac{1}{N} < \epsilon$$

and so we are done. □

### 1.1.2 Some properties of a convergent sequence

**Theorem 1.1** (Uniqueness of Limits). *Let  $\{a_n\}$  be a convergent sequence. Then the limit of the sequence is unique.*

**Definition 1.3.** A sequence  $\{a_n\}$  is *bounded above* if there exists  $M \in \mathbb{R}$  such that  $a_n < M$  for all  $n \in \mathbb{N}$ . Similarly, a sequence  $\{a_n\}$  is *bounded below* if there exists  $m \in \mathbb{R}$  such that  $a_n > m$  for all  $n \in \mathbb{N}$ . A sequence  $\{a_n\}$  is *bounded* if it is bounded above and below.

**Theorem 1.2** (Boundedness of Convergent Sequences). *If  $\{a_n\}$  is a convergent sequence, then it is bounded.*

**Theorem 1.3** (Arithmetic Properties of Convergent Sequences). *Let  $\{a_n\}$  and  $\{b_n\}$  be two sequences. Suppose  $\lim_{n \rightarrow \infty} a_n = A$  and  $\lim_{n \rightarrow \infty} b_n = B$  for some  $A, B \in \mathbb{R}$ . Then*

$$(a) \lim_{n \rightarrow \infty} (a_n + b_n) = \lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} b_n = A + B$$

$$(b) \lim_{n \rightarrow \infty} (a_n b_n) = \left( \lim_{n \rightarrow \infty} a_n \right) \left( \lim_{n \rightarrow \infty} b_n \right) = AB$$

$$(c) \lim_{n \rightarrow \infty} c a_n = c \lim_{n \rightarrow \infty} a_n = cA \text{ for any } c \in \mathbb{R}.$$

$$(d) \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n} = \frac{A}{B}, \text{ provided that } b_n \neq 0 \text{ for all } n \in \mathbb{N} \text{ and } B \neq 0.$$

**Theorem 1.4** (Sandwich Theorem). *Let  $\{a_n\}$ ,  $\{b_n\}$  and  $\{c_n\}$  be sequences. Suppose that*

$$a_n \leq b_n \leq c_n$$

*for all  $n \in \mathbb{N}$ , and that  $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$ . Then  $\lim_{n \rightarrow \infty} b_n = L$ .*

### 1.1.3 Monotone Sequences

**Definition 1.4.** Let  $\{a_n\}$  be a sequence. We say that  $\{a_n\}$  is *increasing* if  $a_n \leq a_{n+1}$  for all  $n \in \mathbb{N}$  and is *decreasing* if  $a_n \geq a_{n+1}$  for all  $n \in \mathbb{N}$ . We say that  $\{a_n\}$  is a *monotone* if it is either increasing or decreasing.

**Theorem 1.5.** (a) *If  $\{a_n\}$  is increasing and bounded above, then  $\{a_n\}$  converges.*

(b) *If  $\{a_n\}$  is decreasing and bounded below, then  $\{a_n\}$  converges.*

### 1.1.4 Subsequences and the Bolzano-Weiertrass Theorem

**Definition 1.5.** Let  $\{a_n\}$  be a sequence of real numbers, and let  $n_1 < n_2 < n_3 < \dots$  be a strictly increasing sequence of natural numbers. Then the sequence

$$a_{n_1}, a_{n_2}, a_{n_3} \dots$$

is called a subsequence of  $\{a_n\}$  and is denoted by  $\{a_{n_k}\}$ , where  $k \in \mathbb{N}$  indexes the subsequence.

**Theorem 1.6.** *If  $\{a_n\}$  is a sequence converging to  $L$ , then every subsequence  $\{a_{n_k}\}$  also converges to  $L$ .*

**Theorem 1.7.** *Let  $\{a_n\}$  be a sequence. The  $\{a_n\}$  has a monotone subsequence.  $\odot$*

*Proof.* To prove this theorem, we call the  $k$ -th term *dominant* if it is greater than or equal to all the following terms. In other words, the term  $a_k$  is dominant if  $a_k \geq a_m$  for all  $m \geq k$ . There are two cases to consider:

Case 1: There are infinitely many dominant terms in the sequence  $\{a_n\}$ . Then we have a infinite subsequence  $a_{n_1}, a_{n_2}, a_{n_3}, \dots$ . Now  $a_{n_i}$  is dominant for all  $i \in \mathbb{N}$  and  $n_i < n_{i+1}$ , so we must have  $a_{n_i} \geq a_{n_{i+1}}$ . Similarly, we have

$$a_{n_1} \geq a_{n_2} \geq a_{n_3} \geq \dots$$

Hence we have a decreasing subsequence  $a_{n_k}$  and we are done.

Case 2: There are only finitely many dominant terms in the sequence  $\{a_n\}$ . (including the case where there is no dominant term). Suppose  $a_N$  is the last dominant term, and let  $n_1 = N + 1$ . Now  $a_{n_1}$  is not dominant and so there must exist  $n_2 > n_1$  such that  $a_{n_1} \leq a_{n_2}$ . Continuing this way, we obtain an increasing subsequence  $a_{n_1}, a_{n_2}, a_{n_3}, \dots$  as desired.  $\square$

**Theorem 1.8** (the Bolzano-Weiertrass Theorem). *Let  $\{a_n\}$  be a bounded sequence. Then  $\{a_n\}$  has a convergent subsequence.*

### 1.1.5 Cauchy's Convergence Criterion

**Definition 1.6.** A sequence  $\{a_n\}$  is called a Cauchy sequence if  $\forall \epsilon > 0, \exists N \in \mathbb{N}$  such that if  $n, m > N$ , then  $|a_n - a_m| < \epsilon$ .

**Theorem 1.9.** Let  $\{a_n\}$  be a convergent sequence. Then  $\{a_n\}$  is a Cauchy sequence.

**Theorem 1.10.** Let  $\{a_n\}$  be a Cauchy sequence. Then  $\{a_n\}$  is a convergent sequence. ☺

## 1.2 Series

### 1.2.1 Series

**Definition 1.7.** Let  $\{a_n\}$  be a sequence of real numbers. The sequence  $\{s_k\}$  can be defined by

$$s_1 = a_1$$

$$s_2 = a_1 + a_2$$

$$s_3 = a_1 + a_2 + a_3$$

$$\vdots$$

$$s_k = a_1 + a_2 + a_3 + \cdots + a_k = \sum_{n=1}^k a_n$$

is called the sequence of partial sums of the series  $\sum_{n=1}^{\infty} a_n$ .

The term  $a_n$  is called the  $n$ -th term of the series and the term  $s_k$  is called the  $k$ -th partial sum of the series. The series  $\sum_{n=1}^{\infty} a_n$  is said to converge to some number  $L$  if the associated sequence  $\{s_k\}$  of the partial sums converges to  $L$ . We write

$$\sum_{n=1}^{\infty} a_n = L$$

and we refer to  $L$  as the sum of the series. If the sequence  $\{s_k\}$  of partial sums diverges, then we say that the series  $\sum_{n=1}^{\infty} a_n$  diverges.

### 1.2.2 Some tests for convergence

**Theorem 1.11** (Cauchy Convergence Criterion for Series). *The series  $\sum_{n=1}^{\infty} a_n$  converges if and only if for any  $\epsilon > 0$  there exists  $N \in \mathbb{N}$  such that if  $n > m > N$ , then*

$$|a_{m+1} + a_{m+2} + \cdots + a_n| < \epsilon.$$

**Theorem 1.12** (Comparison Test). *Suppose that  $\{a_n\}$  and  $\{b_n\}$  are sequences satisfying  $0 \leq a_n \leq b_n$  for all  $n \in \mathbb{N}$ .*

(a) *If  $\sum_{n=1}^{\infty} b_n$  converges, then  $\sum_{n=1}^{\infty} a_n$  converges*

(b) *If  $\sum_{n=1}^{\infty} a_n$  diverges, then  $\sum_{n=1}^{\infty} b_n$  diverges.*

**Theorem 1.13** (Ratio Test). *Let  $\{a_n\}$  be a sequence with  $a_n > 0$  for all  $n \in \mathbb{N}$ . Suppose that*

$$\lim_{n \rightarrow \infty} \frac{a_n}{a_{n+1}}$$

*exists and equals  $r$ . The series  $\sum_{n=1}^{\infty} a_n$  converges if  $r < 1$  and diverges if  $r > 1$ .*

### 1.2.3 Absolute and Conditional Convergence

**Definition 1.8.** Let  $\{a_n\}$  be a sequence of real numbers. The series  $\sum_{n=1}^{\infty} a_n$  is said to be absolutely convergent if the series  $\sum_{n=1}^{\infty} |a_n|$  is convergent. The series  $\sum_{n=1}^{\infty} a_n$  is said to be conditionally convergent if it converges but does not converge absolutely, that is,  $\sum_{n=1}^{\infty} a_n$  converges, but  $\sum_{n=1}^{\infty} |a_n|$  diverges.

## 2 Sequences and Series of Functions

We will now discuss the convergence of sequences and series of real-valued functions defined on a set  $X$ . In most cases,  $X$  would be a subset of  $\mathbb{R}$ . Since we are dealing with the convergence problems of sequences and series, we will naturally be interested in knowing if the limit functions, whenever they exist, will preserve properties such as boundedness, continuity, uniform continuity, differentiability and integrability. *In other words, if the sequence of functions under consideration is **bounded, continuous, differentiable or integrable**, and if the limit function of the sequence exists, does it carry these properties?*

### 2.1 Pointwise convergence of sequences of functions

**Definition 2.1.** Suppose  $\{f_n\}$ ,  $n = 1, 2, 3, \dots$  is a sequence of functions defined on a set  $E$ , and suppose that the sequence of numbers  $\{f_n(x)\}$  converges for every  $x \in E$ . We can define a function  $f$  by

$$f(x) = \lim_{n \rightarrow \infty} f_n(x), x \in E.$$

Under these circumstances, we say that  $\{f_n\}$  converges on  $E$  and that  $f$  is the *limit*, or the *limit function*, of  $\{f_n\}$ . Sometimes we shall use a more descriptive terminology and shall say that " $\{f_n\}$  converges for every  $x \in E$ ", and if we define

$$f(x) = \lim_{n \rightarrow \infty} \sum_{i=1}^n f_i(x), x \in E,$$

then the function  $f$  is called the *sum* of the series  $\sum f_n$ .

*Note 2.1.* The main problem which arises is to determine whether important properties of functions are preserved under the limit operations mentioned above. For instance, if the functions  $\{f_n\}$  are bounded, continuous, differentiable, or integrable, is the same true of the

limit function? What are the relations between  $f'_n$  and  $f'$ , say, or between the integrals of  $f_n$  and that of  $f$ ?

To say that  $f$  is continuous at a limit point  $x$  means

$$\lim_{t \rightarrow x} f(t) = f(x).$$

Hence, to ask whether the limit of a sequence of continuous functions is continuous is the same as to ask whether

$$\lim_{t \rightarrow x} \lim_{n \rightarrow \infty} f_n(t) = \lim_{n \rightarrow \infty} \lim_{t \rightarrow x} f_n(t),$$

i.e., whether the order in which limit processes are carried out is immaterial.

We shall now show by means of several examples that limit processes cannot in general be interchanged without affecting the result. Afterward, we shall prove that under certain conditions the order in which limit operations are carried out is immaterial.

**Example 2.1.** For  $m = 1, 2, 3, \dots$ ,  $n = 1, 2, 3, \dots$ , let

$$s_{m,n} = \frac{m}{m+n}.$$

Then, for every fixed  $n$ ,

$$\lim_{m \rightarrow \infty} s_{m,n} = 1,$$

so that

$$\lim_{n \rightarrow \infty} \lim_{m \rightarrow \infty} s_{m,n} = 1.$$

On the other hand, for every fixed  $m$ ,

$$\lim_{n \rightarrow \infty} s_{m,n} = 0,$$

so that

$$\lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} s_{m,n} = 0.$$

**Example 2.2.** Let

$$f_n(x) = \frac{x^2}{(1+x^2)^n}, \quad x \in \mathbb{R}; n = 0, 1, 2, \dots,$$

and consider

$$f(x) = \sum_{n=0}^{\infty} f_n(x) = \sum_{n=0}^{\infty} \frac{x^2}{(1+x^2)^n}.$$

Since  $f_n(0) = 0$ , we have  $f(0) = 0$ . For  $x \neq 0$ , the last series in the above equation is a convergent geometric series with sum  $1 + x^2$ . Hence

$$f(x) = \begin{cases} 0 & \text{if } x = 0, \\ 1 + x^2 & \text{if } x \neq 0, \end{cases}$$

so that a convergent series of continuous functions may have a discontinuous sum.

*Note 2.2.* After these examples, which show what can go wrong if the limit processes are interchanged carelessly, we will define a new mode of convergence, stronger than point wise convergence as defined in *Definition 2.1*, which will enable us to arrive at positive results.

## 2.2 Uniform convergence of sequences of functions

**Definition 2.2.** We say that a sequence of functions  $\{f_n\}$ ,  $n = 1, 2, 3, \dots$ , converges *uniformly* on  $E$  to a function  $f$  if for every  $\epsilon > 0$  there exists  $N \in \mathbb{N}$  such that  $n \geq N$  implies

$$|f_n(x) - f(x)| \leq \epsilon$$

for all  $x \in E$ .

**Theorem 2.1** (Cauchy Critirion ☺). *The sequence of functions  $\{f_n\}$ , defined on  $E$ , converges uniformly on  $E$ , if and only if for every  $\epsilon > 0$  there exists an integer  $N$  such that  $m \geq N, n \geq N, x \in E$  implies*

$$|f_m(x) - f_n(x)| \leq \epsilon.$$

*Proof.* Assume that  $f_n \rightarrow f$  uniformly on  $E$ . Then, given  $\epsilon > 0$ , we can find  $N$  such that  $n > N$  implies  $|f_n(x) - f(x)| < \frac{\epsilon}{2}$  for all  $x \in E$ .

Taking  $m > N$ , we also have  $|f_m(x) - f(x)| < \frac{\epsilon}{2}$ . Thus it follows from the *triangular inequality* that

$$|f_n(x) - f_m(x)| < |f_n(x) - f(x)| + |f(x) - f_m(x)| < \epsilon.$$

Conversely, suppose that the Cauchy Criterion is satisfied. Then, for each  $x \in E$ , the sequence  $\{f_n(x)\}$  converges. Let  $f(x) = \lim_{n \rightarrow \infty} f_n(x)$  for all  $x \in E$ . We have to prove that the convergence is uniform.

If  $\epsilon > 0$  is given, we can choose an  $N$  such that the Cauchy Criterion implies that

$$\forall n > N, k = 1, 2, 3, \dots, |f_n(x) - f_{n+k}(x)| < \frac{\epsilon}{2}$$

for all  $x \in E$ . Therefore

$$\lim_{k \rightarrow \infty} |f_n(x) - f_{n+k}(x)| = |f_n(x) - f(x)| \leq \frac{\epsilon}{2}$$

for all  $x \in E$ . Hence,  $n > N$  implies  $|f_n(x) - f(x)| < \epsilon$  for all  $x \in E$ . This proves that  $f_n \rightarrow f$  uniformly in  $E$ . □

**Theorem 2.2.** *Suppose*

$$\lim_{n \rightarrow \infty} f_n(x) = f(x) \quad (x \in E).$$

*Put*

$$M_n = \sup_{x \in E} |f_n(x) - f(x)|.$$

*Then  $f_n \rightarrow f$  uniformly on  $E$  if and only if  $M_n \rightarrow 0$  as  $n \rightarrow \infty$ .*

**Example 2.3.** *Let  $\{f_n\}$  be a sequence of functions defined on  $\mathbb{R}$  such that  $f_n(x) = \frac{x^{2n}}{1 + x^{2n}}$  for all  $x \in \mathbb{R}, n \in \mathbb{N}$ . Prove that  $\{f_n\}$  converges pointwise but not uniformly on  $\mathbb{R}$ .*

*Proof.* Since  $\lim_{n \rightarrow \infty} f_n(x)$  exists for all  $x \in \mathbb{R}$ , according to pointwise convergence, the limit function  $f$  is given by

$$f(x) = \begin{cases} 0 & \text{if } |x| < 1 \\ \frac{1}{2} & \text{if } |x| = 1 \\ 1 & \text{if } |x| > 1 \end{cases}$$

Put

$$M_n = \sup_{x \in E} |f_n(x) - f(x)|.$$

In this case, we have  $M_n = \frac{1}{2}$ . Thus, it follows from Theorem 2.2 that  $\{f_n\}$  does not converge to  $f$  uniformly on  $\mathbb{R}$ .  $\square$

### 2.2.1 Uniform convergence and boundedness

**Theorem 2.3.** *If  $\{f_n\}$  is a sequence of bounded functions on  $E$ , and if  $f_n \rightarrow f$  uniformly on  $E$ , then  $f$  is bounded on  $E$ .*

*Proof.* Let  $\epsilon = 1$ . Since  $f_n \rightarrow f$  uniformly on  $E$ ,  $\exists n \in \mathbb{N}$  such that  $n > N$  implies

$$|f_n(x) - f(x)| < 1$$

for all  $x \in E$ . Particularly, consider  $n = N + 1$ . It follows that

$$|f_{N+1}(x) - f(x)| < 1$$

for all  $x \in E$ , that is

$$|f(x)| < |f_{N+1}(x)| + 1$$

for all  $x \in E$ . Since  $f_{N+1}$  is bounded on  $E$ , there exists  $B > 0$  such that

$$|f_{N+1}(x)| < B$$

for all  $x \in E$ . Thus,

$$|f(x)| < B + 1$$

for all  $x \in E$ . This proves that  $f$  is bounded on  $E$ .  $\square$

**Example 2.4.** *A sequence of bounded functions that are pointwise convergent, but the limit function is not bounded.*

$$\text{Consider } f_n(x) = \begin{cases} n & \text{if } x \in (0, \frac{1}{n}] \\ \frac{1}{x} & \text{if } x \in (\frac{1}{n}, 1]. \end{cases}$$

*The limit function  $f$  is given by*

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) = \frac{1}{x}$$

*for all  $x \in (0, 1]$ .*

*Since  $\forall n \in \mathbb{N}$ ,  $|f_n(x)| < n + 1$  for all  $x \in (0, 1]$ , all the functions in the sequence are bounded. However,  $f(x) \rightarrow \infty$  as  $x \rightarrow 0^+$ , the limit function  $f$  is not bounded on  $(0, 1]$ .*

### 2.2.2 Uniform convergence and continuity

**Theorem 2.4** ( $\odot$ ). *Suppose  $f_n \rightarrow f$  uniformly on a set  $E$ . Let  $x$  be a limit point of  $E$ , and suppose that*

$$\lim_{t \rightarrow x} f_n(t) = A_n \quad (n = 1, 2, 3, \dots).$$

*Then  $\{A_n\}$  converges, and*

$$\lim_{t \rightarrow x} f(t) = \lim_{n \rightarrow \infty} A_n.$$

*In other words, the conclusion is that*

$$\lim_{t \rightarrow x} \lim_{n \rightarrow \infty} f_n(t) = \lim_{n \rightarrow \infty} \lim_{t \rightarrow x} f_n(t)$$

*Proof.* Let  $\epsilon > 0$  be given. By the uniform convergence of  $\{f_n\}$ , there exists  $N$  such that  $n \geq N, m \geq N, t \in E$  imply

$$|f_n(t) - f_m(t)| \leq \epsilon.$$

Letting  $t \rightarrow x$ , we obtain

$$|A_n - A_m| \leq \epsilon$$

for  $n \geq N$ ,  $m \geq N$ , so that  $\{A_n\}$  is a Cauchy sequence and therefore converges, say to  $A$ .  
Next,

$$|f(t) - A| \leq |f(t) - f_n(t)| + |f_n(t) - A_n| + |A_n - A|.$$

We first choose  $n$  such that

$$|f(t) - f_n(t)| \leq \frac{\epsilon}{3}$$

for all  $t \in E$  (this is possible by the uniform convergence), and such that

$$|A_n - A| \leq \frac{\epsilon}{3}.$$

Then, for this  $n$ , we choose a neighbourhood  $V$  of  $x$  such that

$$|f_n(t) - A_n| \leq \frac{\epsilon}{3}$$

if  $t \in V \cap E$ ,  $t \neq x$ . Substituting the inequalities above, we see that

$$|f(t) - A| \leq \epsilon,$$

provided that  $t \in V \cap E$ ,  $t \neq x$ . This is equivalent to  $\lim_{t \rightarrow x} f(t) = \lim_{n \rightarrow \infty} A_n$ . □

**Theorem 2.5.** *If  $\{f_n\}$  is a sequence of continuous functions on  $E$ , and if  $f_n \rightarrow f$  uniformly on  $E$ , then  $f$  is continuous on  $E$ .*

*Note 2.3.* If  $x_0$  is an accumulation point of  $E$ , then  $f$  is continuous at  $x = x_0$  if and only if

$$\lim_{x \rightarrow x_0} \lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \lim_{x \rightarrow x_0} f_n(x).$$

If  $x_0$  is an isolated point of  $E$ , then  $f$  is automatically continuous at  $x = x_0$ .

Thus, proof of Theorem 2.5 follows directly from Theorem 2.4.

*Note 2.4.* Is the converse of Theorem 2.5 true?

The converse is not true in general, i.e. uniform convergence is a sufficient but not necessary condition to transmit continuity from individual terms to the limit function.

In Example 2.5, we will see a non-uniformly convergent sequence of continuous functions with a continuous limit function.

**Example 2.5.** Consider  $\{f_n\}$  defined on  $[0, 1]$  such that  $f_n(x) = n^2 x(1 - x)^n$  for all  $n \in \mathbb{N}$ ,  $x \in [0, 1]$ . The limit function  $f$  exists and

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} n^2 x(1 - x)^n = 0$$

for all  $x \in [0, 1]$ .

Obviously, each term in the sequence, as well as the limit function, is continuous on  $[0, 1]$ .

However, Let  $M_n = \sup_{x \in [0, 1]} |f_n(x) - f(x)|$ . Since  $\lim_{n \rightarrow \infty} M_n > 0$ , it follows from Theorem 2.2 that  $\{f_n\}$  does not converge to  $f$  uniformly on  $[0, 1]$ .

### 2.2.3 Uniform convergence and Riemann integration

**Theorem 2.6** ( $\odot$ ). Suppose  $f_n \in \mathcal{R}([a, b])$  for  $n = 1, 2, 3, \dots$  and suppose  $f_n \rightarrow f$  uniformly on  $[a, b]$ . Then  $f \in \mathcal{R}([a, b])$ , and

$$\int_a^b f = \lim_{n \rightarrow \infty} \int_a^b f_n.$$

(The existence of the limit is part of the conclusion.)

*Proof.* It suffices to prove this for real  $f_n$ . Put

$$\epsilon_n = \sup |f_n(x) - f(x)|,$$

the supremum being taken over  $a \leq x \leq b$ . Then

$$f_n - \epsilon_n \leq f \leq f_n + \epsilon_n,$$

so that the upper and lower integrals of  $f$  satisfy

$$\int_a^b (f_n - \epsilon_n) \leq \underline{\int} f \leq \overline{\int} f \leq \int_a^b (f_n + \epsilon_n).$$

Hence

$$0 \leq \overline{\int} f - \underline{\int} f \leq 2\epsilon_n(b - a).$$

Since  $\epsilon_n \rightarrow 0$  as  $n \rightarrow \infty$ , the upper and lower integrals of  $f$  are equal.

Thus  $f \in \mathcal{R}([a, b])$ . Another application of the inequality above now yields

$$\left| \int_a^b f - \int_a^b f_n \right| \leq \epsilon_n(b-a).$$

This implies  $\int_a^b f = \lim_{n \rightarrow \infty} \int_a^b f_n$ . □

**Note 2.5 (Riemann integrability).** By  $f \in \mathcal{R}([a, b])$ , we mean that the function  $f$  is Riemann integrable on  $[a, b]$ . The definition is as follows:

Let  $f : [a, b] \rightarrow \mathbb{R}$  be a bounded (not necessary continuous) function on  $[a, b]$ . Then  $f$  is said to be *Riemann integrable* on  $[a, b]$  if  $U(f) = L(f)$ , where

$$U(f) = \inf \{U(f, P) : P \text{ is a partition of } [a, b]\}$$

$$L(f) = \sup \{L(f, P) : P \text{ is a partition of } [a, b]\}.$$

In this case, the common value of  $U(f)$  and  $L(f)$  is called the Riemann integral of  $f$  on  $[a, b]$ , and is denoted by  $\int_a^b f(x)dx$ .

### 2.2.4 Uniform convergence and differentiation

By analogy with Theorem 2.3, 2.4, and 2.6, one might expect the following result to hold:

**Theorem 2.7 (☉).** Suppose that  $\{f_n\}$  is a sequence of functions, differentiable on  $[a, b]$  and such that  $\{f_n(x_0)\}$  converges for some point  $x_0$  on  $[a, b]$ . If  $\{f_n\}$  converges uniformly on  $[a, b]$ , then  $\{f'_n\}$  converges uniformly on  $[a, b]$ , to a function  $f'$ , and

$$f'(x) = \lim_{n \rightarrow \infty} f'_n(x) \quad (a \leq x \leq b).$$

Actually, in Example 2.6, we will see that this cannot be true.

**Example 2.6.** A sequence of differentiable functions  $\{f_n\}$  with limit 0 for which  $\{f'_n\}$  diverges.

Let  $f_n(x) = \frac{\sin nx}{\sqrt{n}}$  if  $x \in \mathbb{R}$ ,  $n = 1, 2, 3, \dots$

Then  $\lim_{n \rightarrow \infty} f_n(x) = 0$  for all  $x \in \mathbb{R}$ .

But  $f'_n(x) = \sqrt{n} \cos nx$ , so  $\lim_{n \rightarrow \infty} f'_n(x)$  does not exist for any  $x \in \mathbb{R}$ .

**Theorem 2.8** ( $\odot$ ). Assume that each term of  $\{f_n\}$  is a real-valued function having a finite derivative at each point of an open interval  $(a, b)$ . Assume that for at least one point  $x_0$  in  $(a, b)$  the sequence  $\{f_n(x_0)\}$  converges. Assume further that there exists a function  $g$  such that  $f'_n \rightarrow g$  uniformly on  $(a, b)$ . Then:

a) There exists a function  $f$  such that  $f_n \rightarrow f$  uniformly on  $(a, b)$ .

b) For each  $x$  in  $(a, b)$  the derivative  $f'(x)$  exists and equals  $g(x)$ .

*Proof.* Assume that  $c \in (a, b)$  and define a new sequence  $\{g_n\}$  as follows:

$$g_n(x) = \begin{cases} \frac{f_n(x) - f_n(c)}{x - c} & \text{if } x \neq c, \\ f'_n(c) & \text{if } x = c. \end{cases}$$

The sequence  $\{g_n\}$  so formed depends on the choice of  $c$ . Convergence of  $\{g_n(c)\}$  follows from the hypothesis, since  $g_n(c) = f'_n(c)$ . We will prove next that  $\{g_n\}$  converges uniformly on  $(a, b)$ . If  $x \neq c$ , we have

$$g_n(x) - g_m(x) = \frac{h_n(x) - h_m(x)}{x - c},$$

where  $h(x) = f_n(x) - f_m(x)$ . Now  $h'(x)$  exists for each  $x \in (a, b)$  and equals  $f'_n(x) - f'_m(x)$ .

Applying the Mean Value Theorem, we get

$$g_n(x) - g_m(x) = f'_n(x_1) - f'_m(x_1)$$

where  $x_1$  lies between  $x$  and  $c$ . Since  $\{f'_n\}$  converges uniformly on  $(a, b)$ , we can use the Cauchy condition to deduce that  $\{g_n\}$  converges uniformly on  $(a, b)$ .

Now we can show that  $\{f_n\}$  converges uniformly on  $(a, b)$ . Let us form a particular sequence

$\{g_n\}$  corresponding to the special point  $c = x_0$  for which  $\{f_n(x_0)\}$  is assumed to converge.

We can write

$$f_n(x) = g_n(x)(x - x_0) + f_n(x_0)$$

an equation which holds for every  $x \in (a, b)$ . Hence we have

$$f_n(x) - f_m(x) = f_n(x_0) - f_m(x_0) + [g_n(x) - g_m(x)](x - x_0).$$

This equation, with the help of the Cauchy condition, establishes the convergence of  $\{f_n\}$  on  $(a, b)$ . This proves (a).

To prove (b), return to the sequence  $\{g_n\}$  defined for an *arbitrary* point  $c$  in  $(a, b)$  and let  $G(x) = \lim_{n \rightarrow \infty} g_n(x)$ . The hypothesis that  $f'_n$  exists means that  $\lim_{x \rightarrow c} g_n(x) = g_n(c)$ . In other words, each  $g_n$  is continuous at  $c$ . Since  $g_n \rightarrow G$  uniformly on  $(a, b)$ , the limit function  $G$  is also continuous at  $c$ . This means that

$$G(c) = \lim_{x \rightarrow c} G(x),$$

the existence of the limit being part of the conclusion. But, for  $x \neq c$ , we have

$$G(x) = \lim_{n \rightarrow \infty} g_n(x) = \lim_{n \rightarrow \infty} \frac{f_n(x) - f_n(c)}{x - c} = \frac{f(x) - f(c)}{x - c}.$$

Hence, the derivative  $f'(c)$  exists and equals  $G(c)$ . But

$$G(c) = \lim_{n \rightarrow \infty} g_n(c) = \lim_{n \rightarrow \infty} f'_n(c) = g(c);$$

hence  $f'(c) = g(c)$ . Since  $c$  is an arbitrary point of  $(a, b)$ , this proves (b). □

*Note 2.6 (Mean Value Theorem).* Let  $f : [a, b] \rightarrow \mathbb{R}$  be a function which is continuous on  $[a, b]$  and differentiable on  $(a, b)$ . There exists  $c \in (a, b)$  such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

## 2.3 Uniform convergence of series of functions

**Definition 2.3.** Given a sequence  $\{f_n\}$  of functions defined on a set  $S$ . For each  $x$  in  $S$ , let

$$s_n(x) = \sum_{k=1}^n f_k(x) \quad (n = 1, 2, 3, \dots).$$

If there exists a function  $f$  such that  $s_n \rightarrow f$  uniformly on  $S$ , we say the series  $\sum f_n(x)$  converges uniformly on  $S$  and we write

$$\sum_{n=1}^{\infty} f_n(x) = f(x) \quad (\text{uniformly on } S).$$

### 2.3.1 Some tests for uniform convergence of series of functions

**Theorem 2.9** (Cauchy Condition for uniform convergence of series). *The infinite series  $\sum f_n(x)$  converges uniformly if, and only if, for any  $\epsilon > 0$  there is an  $N$  such that  $n > N$  implies*

$$\left| \sum_{k=n+1}^{n+p} f_k(x) \right| < \epsilon \quad \text{for each } p = 1, 2, 3, \dots, \text{ and for every } x \text{ in } S.$$

**Theorem 2.10** (Weierstrass M-test). *Let  $\{M_n\}$  be a sequence of non-negative numbers such that*

$$0 \leq |f_n(x)| \leq M_n, \quad \text{for } n = 1, 2, 3, \dots \text{ and for every } x \text{ in } S.$$

*Then  $\sum f_n(x)$  converges uniformly on  $S$  if  $\sum M_n$  converges.*

### 2.3.2 the Power Series

In this section, we shall derive some properties of functions which are represented by power series, i.e., functions of the form

$$f(x) = \sum_{n=0}^{\infty} c_n x^n$$

or, more generally,

$$f(x) = \sum_{n=0}^{\infty} c_n (x - a)^n. \quad (*)$$

There are called *analytic functions*.

We shall restrict ourselves to real values of  $x$ . Instead of circles of convergence (complex values of  $x$ ), we shall therefore encounter intervals of convergence.

If  $(*)$  converges for  $|x - a| < R$ ,  $f$  is said to be expanded in a power series about the point  $x = a$ . As a matter of convenience, we often take  $a = 0$  without any loss of generality.

### 1. Properties of functions represented by real power series

**Theorem 2.11.** *Suppose the series*

$$\sum_{n=0}^{\infty} c_n x^n \quad (*)$$

*converges for  $|x| < R$ , and define*

$$f(x) = \sum_{n=0}^{\infty} c_n x^n \quad (|x| < R).$$

*Then  $(*)$  converges uniformly on  $[-R + \epsilon, R - \epsilon]$ , no matter which  $\epsilon > 0$  is chosen. The function  $f$  is continuous and differentiable in  $(-R, R)$ , and*

$$f'(x) = \sum_{n=0}^{\infty} n c_n x^{n-1} \quad (|x| < R).$$

**Theorem 2.12.** *Suppose  $\sum c_n$  converges. Put*

$$f(x) = \sum_{n=0}^{\infty} c_n x^n \quad (-1 < x < 1).$$

*Then*

$$\lim_{x \rightarrow 1} f(x) = \sum_{n=0}^{\infty} c_n.$$

**Theorem 2.13.** *suppose*

$$f(x) = \sum_{n=0}^{\infty} c_n x^n,$$

the series converging in  $|x| < R$ . If  $-R < a < R$ , then  $f$  can be expanded in a power series about the point  $x = a$  which converges in  $|x - a| < R - |a|$ , and

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n \quad (|x - a| < R - |a|).$$

*Note 2.7.* This is known as *Taylor's Theorem*.

## 2. Power-series expansions for the exponential and trigonometric functions